
Question 1.1.2 While Euclidean distance is a more commonly known method to measure distance, it's not the only one! Another method is the [Manhattan distance](#). How do we calculate the Manhattan distance between two points? What situation might Manhattan distance be more suitable than Euclidean distance? **(4 points)**

Hint: Consider why it's named *Manhattan distance*—think of a city grid!

$|x_1 - x_2| + |y_1 - y_2|$. Manhattan distance is more suitable than Euclidean distance in situations where movement is restricted to grid-like paths, where you can't only move diagonally.

Question 1.7.1 When doing Knn classification we split our data into training and test sets.

Why do we divide our data into training and test sets? Or in other words what is the point of the training set? What is the point of the test set? Answer both questions. **(7 points)**

Hint: Check out this [section](#) in the textbook.

Training set uses for model learning the pattern of data, test sets uses for evaluating the model by unseen data

Question 1.7.2 Why do we only want to use the test set once? **(3 points)**

it's purpose is to simulate the unseen data, real world data

Question 1.8. Why do we choose k to be an odd number in k -NN? Explain. **(10 points)**

to avoid ties when voting

